

PROJECT REPORT

ON

**“Dynamic Keypad Menu with Threshold Management
of multiple sensors.”**

UNDER THE COURSE

“Operating Systems & Embedded Systems Design”

25EECC304

TEAM DETAILS – TEAM No. 04

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PROBLEM STATEMENT

The objective of this project is to design and implement an RTOS-based dynamic keypad menu system using the ARM Cortex-M3 LPC1768 microcontroller. The system allows the user to set and modify threshold values for temperature, gas concentration, and distance parameters through a 4×4 keypad interface. The threshold entered through the 4×4 keypad for each sensor is used to generate an alert for that respective sensor. All the above-mentioned functionalities should be visualized through LCD display at each and every stage. The system should be implemented to run using a Real-Time Operating System (RTOS).

INTRODUCTION

- Safety monitoring is a critical requirement in industrial, domestic, and laboratory environments where parameters such as distance, gas concentration, and temperature must be continuously observed to prevent hazardous conditions. In this project, an RTOS-based safety monitoring system is implemented using the **ARM Cortex-M3 LPC1768** microcontroller to provide reliable, real-time supervision of these parameters.
- The system integrates an ultrasonic sensor for distance measurement, a gas sensor for detecting harmful gas levels, and a DHT11 sensor for temperature monitoring. A 4×4 keypad is used to allow the user to select the monitoring mode and set safety threshold values dynamically, while a 16×2 LCD displays real-time sensor readings and system status. When any parameter exceeds its predefined threshold, visual alerts using LEDs and an audible buzzer are activated.
- An **RTOS (Real-Time Operating System)** is employed to manage concurrent sensor tasks efficiently, ensuring deterministic operation, proper task synchronization, and safe shared resource access using a mutex for LCD control. This approach improves system responsiveness, scalability, and reliability compared to conventional polling-based designs.

FEATURES SELECTED

- Multi-parameter monitoring of distance, gas concentration, and temperature.
- User-configurable thresholds using a 4×4 keypad for flexibility under different conditions.
- Real-time status display on a 16×2 LCD for continuous feedback.
- Visual and audible alerts using LEDs and a buzzer to indicate unsafe conditions.
- Concurrent task execution using an RTOS to handle multiple sensors efficiently.
- Safe shared resource access using a mutex to protect LCD operations.

ASSUMPTIONS MADE

- The operating environment uses standard atmospheric conditions, assuming a constant speed of sound for ultrasonic distance calculation.
- Only one sensor mode is active at a time, selected through the keypad.
- Gas sensor output voltage is within the ADC input range (0–3.3 V) of the microcontroller.
- The DHT11 sensor provides integer temperature values, sufficient for basic safety monitoring.

JUSTIFICATION FOR EACH DESIGN CHOICE

1.LPC1768 Microcontroller

Chosen for its ARM Cortex-M3 core, built-in 12-bit ADC, multiple GPIOs, timer peripherals, and RTOS support, making it suitable for real-time embedded applications.

2.Ultrasonic Sensor

Provides non-contact distance measurement, suitable for obstacle or proximity-based safety detection.

3. Gas Sensor (Analog Type)

Enables detection of harmful gases through analog voltage variation, easily interfaced using the microcontroller's ADC.

4. DHT11 Temperature Sensor

Selected for its digital output, simple single-wire interface, and ease of integration for basic temperature monitoring.

5.4×4 Keypad

Allows menu-driven user interaction and dynamic threshold setting without requiring external devices.

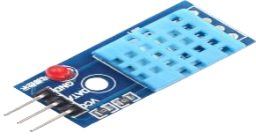
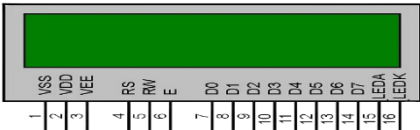
6. 16×2 LCD

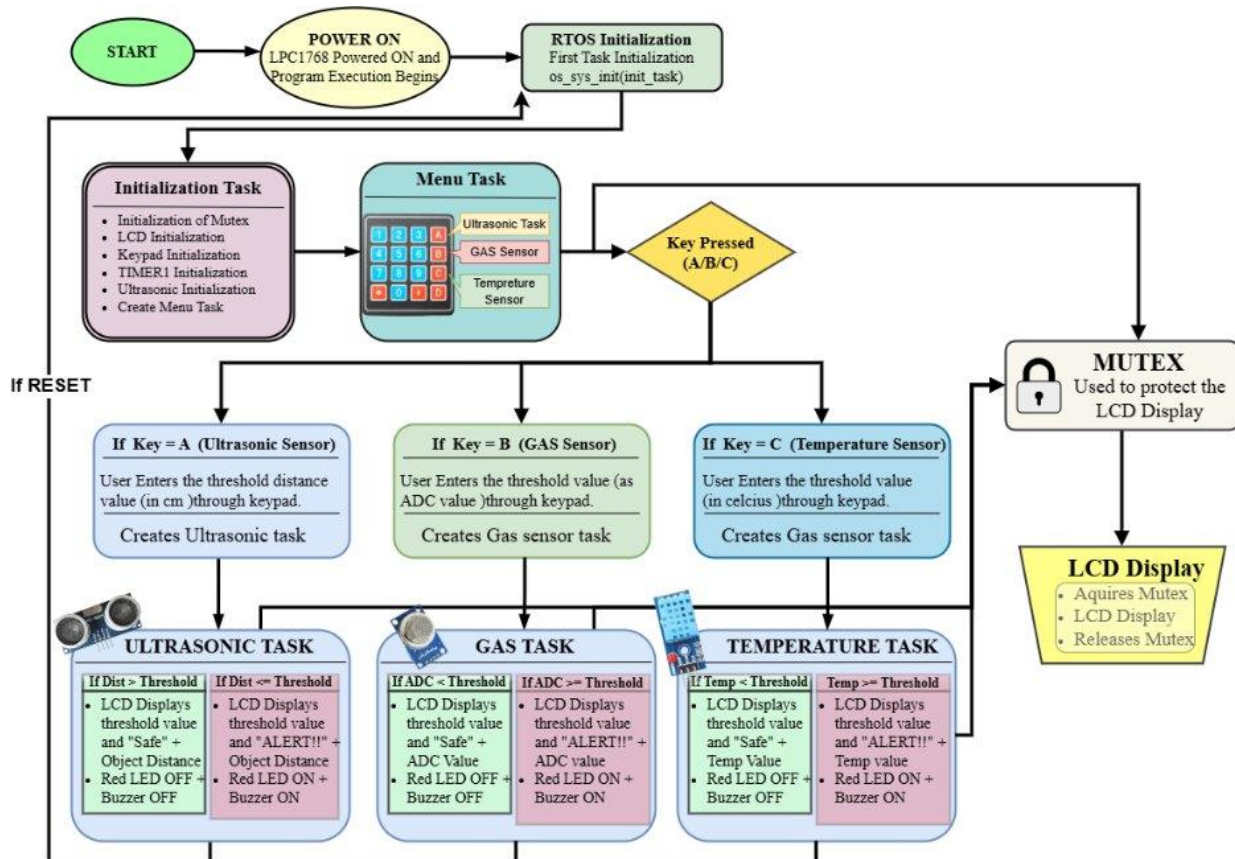
Provides a low-cost and effective interface to display sensor values, thresholds, and system status.

7. LEDs and Buzzer

Offer immediate visual and audible alerts, improving user awareness during unsafe conditions.

PERIPHERALS CHOSEN FOR PROTOTYPING

SL. NO	PERIPHERAL	PERIPHERAL IMAGE
1.	LPC1768 Microcontroller	
2.	Ultrasonic sensor	
3.	Gas sensor	
4.	4X4 matrix keypad	
5.	Temperature sensor	
6.	16x2 LCD	



7.

LEDs & Buzzer



FLOW CHART

Specifications of All Peripherals Used & Calculations

1. Microcontroller – LPC1768 (ARM Cortex-M3)

Specifications:

- Core: ARM Cortex-M3
- Operating voltage: 3.3 V
- Clock frequency: Up to 100 MHz
- GPIO ports: GPIO0 – GPIO4
- ADC resolution: 12-bit
- Timers: 32-bit Timer0–Timer3



- RTOS support: Keil RTX



2. ULTRASONIC SENSOR

Specifications

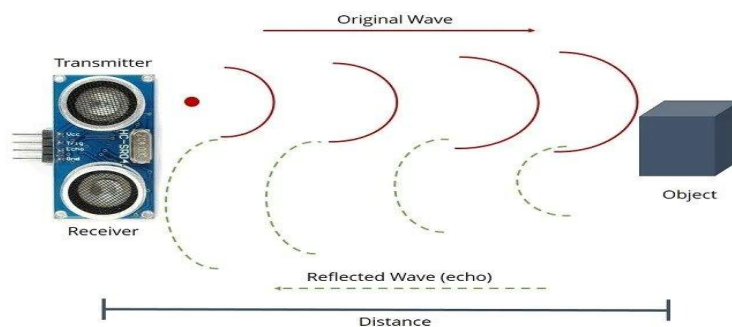
- Operating voltage: 5 V
- Operating frequency: 40 kHz
- Measuring range: 2 cm – 400 cm
- Interface pins: TRIG, ECHO

GPIO Pins Used

- TRIG → P0.0 (Output)
- ECHO → P0.1 (Input)

Ultrasonic Function:

- Sends an ultrasonic pulse.
- Measures echo time.
- Convert time to distance



DISTANCE CALCULATION

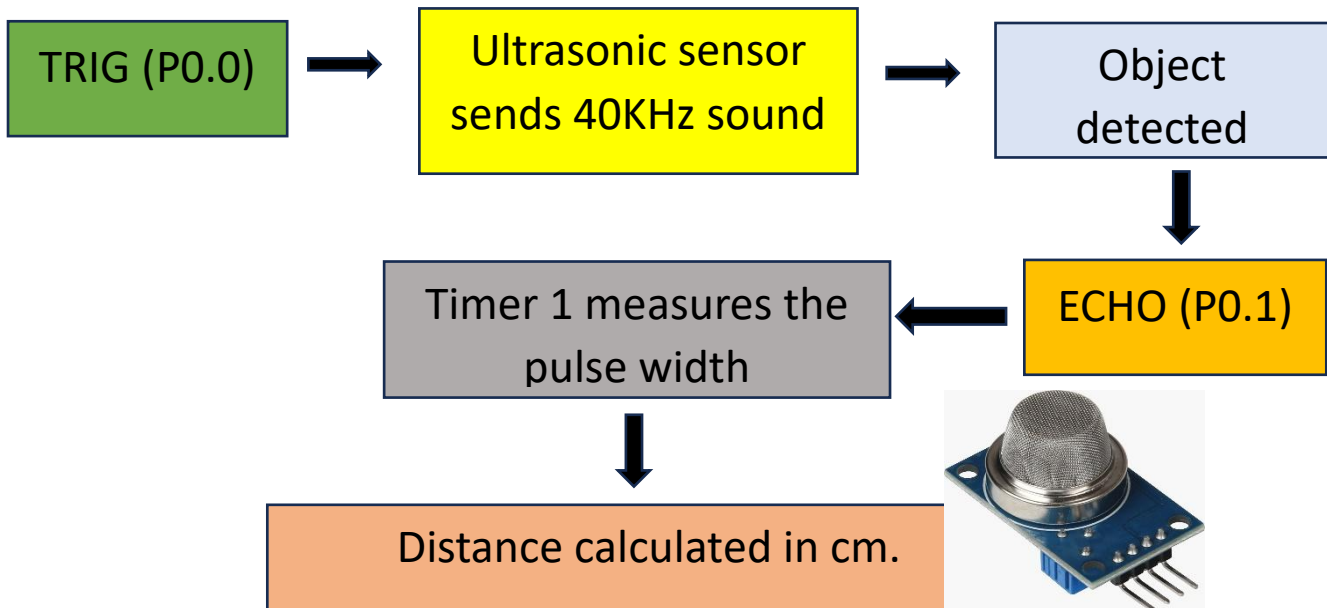
$$\text{Distance (cm)} = \frac{\text{Echo time (t in } \mu\text{s)} \times 0.034}{2}$$

- Speed of sound = 340 m/s = 0.034 cm/s
- We divide by 2 because the sound travels to the object and back to the sensor.

- Formula used in code:

$$\text{Distance(cm)} = \frac{t \times 34}{2000}$$

SIMPLE BLOCK FLOW



3. GAS SENSOR

➤ Specifications

- Operating voltage: 5 V
- Output type: Analog voltage
- Detectable gases: Smoke, LPG etc
- Output range: 0 – 3.3 V

➤ ADC Configuration

- ADC channel: AD0.0
- GPIO pin: P0.23
- Resolution: 12-bit
- ADC range: 0 – 4095

➤ How gas sensing works?

- Inside the gas sensor it contains the heated metal-oxide (SnO_2) element.

- When the sensor detects any smoke or gas the sensor resistance decreases which increases the voltage.
 - This analog voltage is converted to digital value using the ADC.
- Higher gas concentration = Higher output voltage.

ADC CONVERSION CALCULATION

$$\text{ADC value} = \frac{V_{in}}{V_{ref}} \times (2^{12}-1)$$

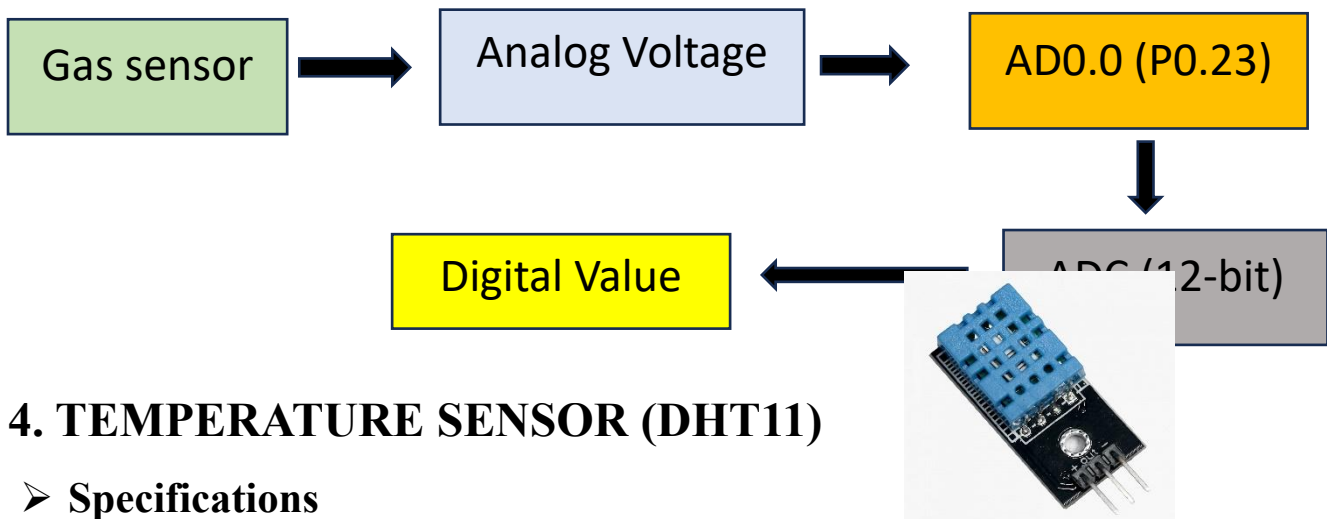
Where,

- V_{in} = Gas sensor voltage
 - V_{ref} = Input voltage to sensor (3.3 v)
 - We take 2^{12} because the ADC is 12-bit.
- Example: Consider $V_{in} = 1.65 \text{ v}$ then,

$$\text{ADC value} = \frac{1.65}{3.3} \times 4095$$

$$\approx 2047$$

SIMPLE BLOCK FLOW



4. TEMPERATURE SENSOR (DHT11)

➤ Specifications

- Operating voltage: 3.3 V - 5 V
- Temperature range: 0°C – 50°C
- Resolution: 1°C
- Output type: Digital value

GPIO Pins Used

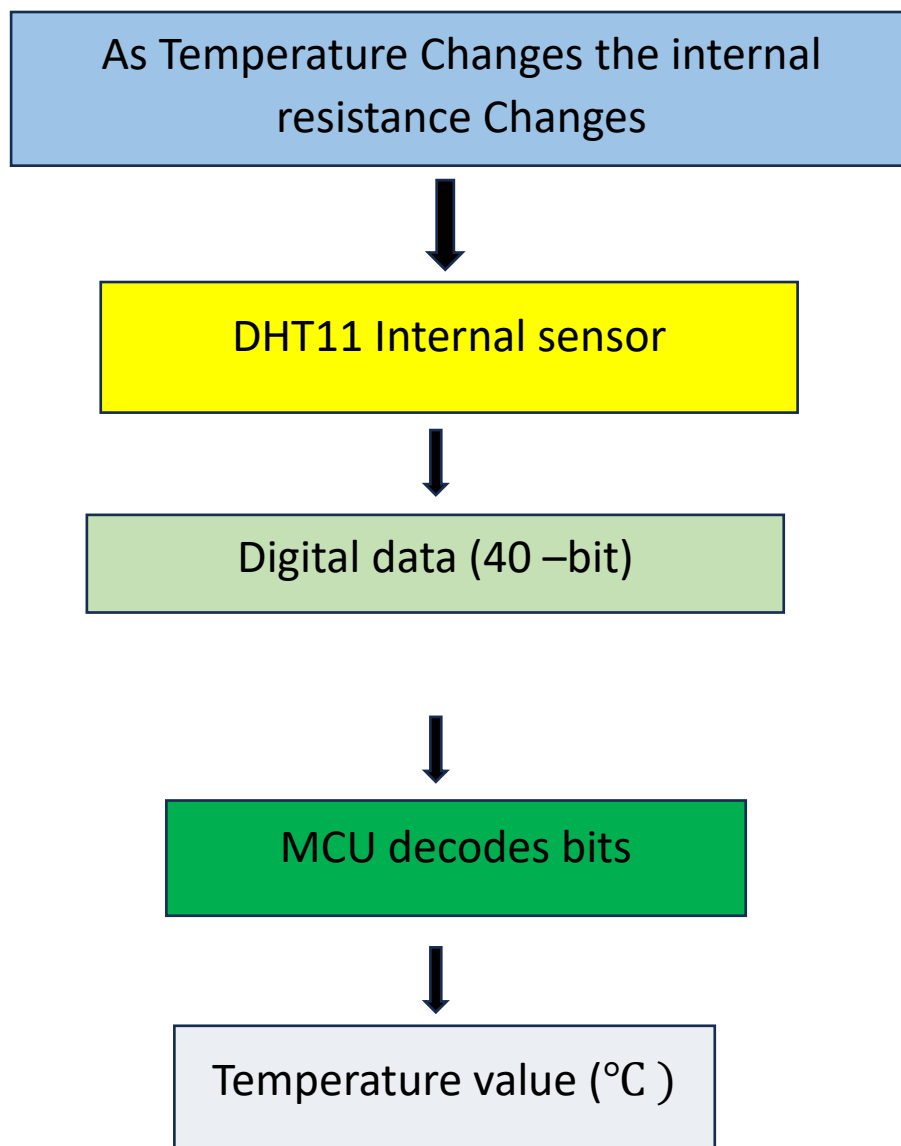
- DATA → P0.16 (Bi-directional GPIO)

➤ Temperature Measurement:

- DHT11 internally uses an NTC thermistor
- As Resistance ↓ as temperature ↑
- Internal ADC converts temperature to digital data
- MCU reads 40-bit data frame
- Temperature extracted directly in °C

NOTE: No external calculation required.

SIMPLE BLOCK FLOW



5. LCD DISPLAY

➤ Specifications

- Operating voltage: 5 V
- Interface mode: 4-bit
- Display: 16 characters × 2 lines



➤ GPIO Pins Used

- RS (Register Select) → P3.25
- RW (Read / Write) → P3.26
- E (Enable) → P4.28
- D4–D7 (Data Bus) → P1.20 – P1.23

6. LEDs & BUZZER

➤ Specifications

- LEDs and Buzzer
- Current: 10–15 mA
- Buzzer voltage: 5 V



➤ GPIO Pins Used:

- Ultrasonic RED LED → P0.25
- Gas Sensor RED LED → P0.17
- Temperature Sensor RED LED → P0.15
- Buzzer → P0.24

7. 4×4 Matrix KEYPAD

➤ Specifications

- Keys: 16
- Interface type: Matrix scanning
- Operating voltage: 3.3 V

➤ GPIO Pins Used:

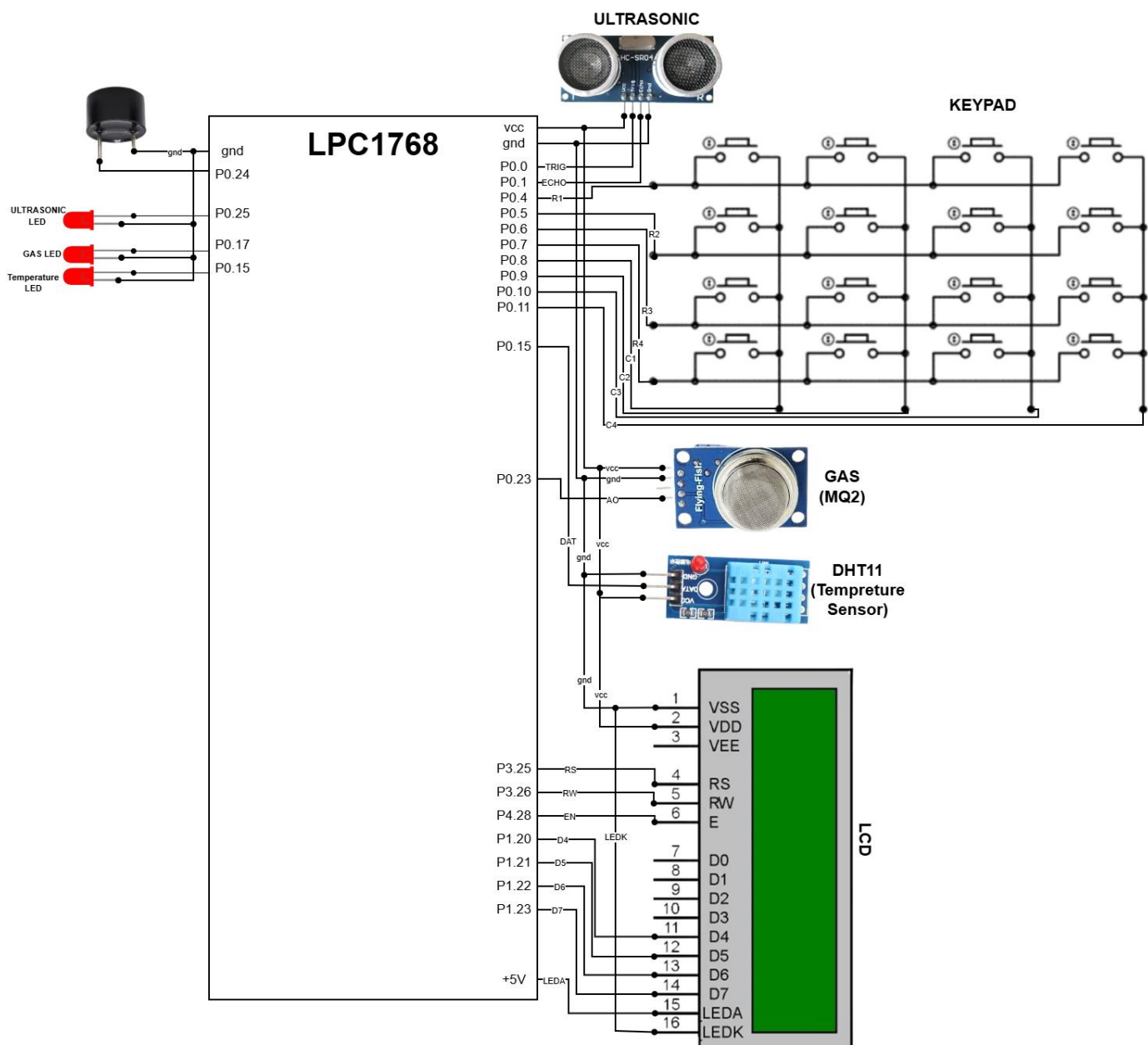
- Rows (Output): P0.4 – P0.7
- Columns (Input): P0.8 – P0.11



➤ Keypad Scanning Logic

- Rows driven LOW one at a time
- Columns monitored for LOW
- Row + Column → Key identification

INTERFACING DIAGRAM



RESULTS

PROTOTYPE IMPLEMENTED



WORKING VIDEO

[Click Here](#)

for the working video of our final prototype.

MEMORY ANALYSIS

- Code memory = 7644 Bytes (Flash Memory)
- RO-Data (memory used by strings and look-up table) = 296 Bytes (Flash Memory)
- RW-Data (memory used by Initialized globals) = 56 Bytes (RAM Memory)
- ZI-Data (Stack + Heap) = 2816 bytes (RAM)
- Total stack size = 200 bytes (50 byte for each task)

TIMING ANALYSIS

- Delay Calculation:
- Ultrasonic Task Delay (for 1 cycle):
 - Task delay = 10ms (Tick time) * 50 = 500ms
 - Ultrasonic Read = 20ms
 - LCD Update (with mutex wait) < 1ms

- Total = 521ms
- GAS Task Delay (for 1 cycle):
 - Task delay = 10ms (Tick time) * 50 = 500ms
 - GAS Read (ADC Conversion) < 1ms
 - LCD Update (with mutex wait) < 1ms
 - Total = 502ms
- Temperature Task Delay (for 1 cycle):
 - Task delay = 10ms (Tick time) * 50 = 500ms
 - DHT read < 25ms
 - LCD Update (with mutex wait) < 1ms
 - Total = 526ms

Total Delay (Worst Case) $\approx$ 525ms

CONCLUSION

- Accurate timing control was required for ultrasonic sensor and DHT11 communication.
- Concurrent LCD access from multiple tasks caused display corruption initially, which was resolved using a mutex.
- Managing multiple peripherals simultaneously under RTOS constraints required proper task structuring and synchronization.
- Debugging real-time behavior and task execution order was challenging during integration.
- Understanding each peripheral working and implementing proper RTOS Kernel object to it.
- Managing GPIO pin conflicts while interfacing multiple peripherals on the same port required careful pin planning.
- Handling time-critical communication without affecting RTOS task scheduling required careful delay and timer usage.

SUMMARY OF OUTCOMES

- Successfully designed and implemented a real-time embedded system using LPC1768 and RTX RTOS.
- Integrated multiple sensors (Ultrasonic, Gas, DHT11) with actuators (LEDs, buzzer) for real-time monitoring.
- Implemented multitasking using RTX, ensuring concurrent sensor operation
- Ensured safe shared resource access by implementing mutex-based synchronization for LCD operations.
- Designed a menu-driven user interface using keypad and LCD for dynamic threshold configuration.
- Validated the complete system through successful hardware testing and result analysis.

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